

What is the problem?

Due date: 12-10-2025

State the Inverse problem. Be specific.

In a valley glacier, the ice flow has two components: 1) internal deformation of ice following the flow law, 2) sliding at the glacier bed, called basal motion. The forward problem calculates the glacier's surface velocity given the glacier's geometry, flow law, and basal motion. The corresponding inverse problem is to estimate the basal motion from observed surface velocities and glacier geometry.

Why is this a problem?

Basal motion is a major contributor to glacier flow, but it is extremely difficult to measure directly. It has been linked to quantities like the amount of water present glacier bed, the properties of the underlying bedrock, which themselves are difficult access through methods like borehole drilling. Surface velocities can be measured more easily using GPS, satellite imagery, or time-lapse photography.

Inversion Algorithm

26-10-2025

Inputs: Basal velocity profile and constant values in the forward model. Compute surface velocity then add noise to it this is input for the inverse model.

Loop: Generate a gaussian kernel for integral operator; discretize the integration; define matrix for Tikonov first-order regularization.

Computing: Generate the solution of the inverse model.

Test stopping criteria: estimated basal velocities from the inverse model and corresponding surface velocity using forward model.

Discuss the Methods

Due date: 19-10-2025

For an ice slab of infinite extent on an inclined plane, the deformational surface velocity can be derived using the force balance and the flow/deformation law (deformation is somewhere between the two extremes of elastic deformation and perfectly plastic deformation) as given below:

$u_c = \frac{2A}{n+1} (\rho g f \sin \alpha)^{n+1} h^{n+1}$ where α is the slope, h is the ice thickness and A, n, ρ and f are constants. Now this is purely deformational velocity along the center flow line of the ice slab. The surface velocity is the sum of this deformational velocity and the velocity due to sliding motion of glacier at the base or basal motion. $u_s = u_c + u_b$

Theoretical models

The above definition of surface velocity says that at a given point on a glacier central flowline, surface velocity is the sum of deformational and basal velocity at that point. A more realistic model would be where surface velocity at a point is a function of deformational and basal velocities at that point and also on the velocity surrounding points in it written in the form of weighted average:

$\ln u_s(x) = \int_0^L g(x'-x) \ln(u_c(x') + u_b(x')) dx'$ Here L is length of the ice slab, and l is an averaging lengthscale. logarithm is taken to linearize the problem as velocity is initially a nonlinear function of slope and ice thickness.

$g(x'-x) = C e^{-\alpha |x'-x|}$
 $\Rightarrow \ln u_s(x) = \int_0^L g(x'-x) \ln u_c(x') dx' = \int_0^L g(x'-x) \ln(1 + \frac{u_b(x')}{u_c(x')}) dx'$

Let L.H.S be denoted by $D(x)$ and $m(x') = \ln(1 + \frac{u_b(x')}{u_c(x')})$

The forward linear problem can be defined as:

$D(x) = \int_0^L g(x'-x) m(x') dx'$ This is the problem of a continuous model with discrete data.

$d_j = \sum_{i=1}^M g_j(x_i) m(x_i) \Delta x$ This can further be described as a discrete linear problem of the form $d = Gm$ by integrating numerically. Note that this is an underdetermined problem and thus we use regularization. Tikonov regularization of first order is used instead of minimizing m we minimize its derivative, this also ensures smooth velocity solutions and is also physically valid as velocity varies smoothly.

Solution and Discussion

Due date: 02-11-2025

Figure 1:

The dotted line shows the basal velocity which is used as an input in the forward model to generate surface velocity. The center flowline velocity is taken as a unit constant. Notice that the calculated surface velocity is smoothened out due to the averaging operator.

Figure 2:

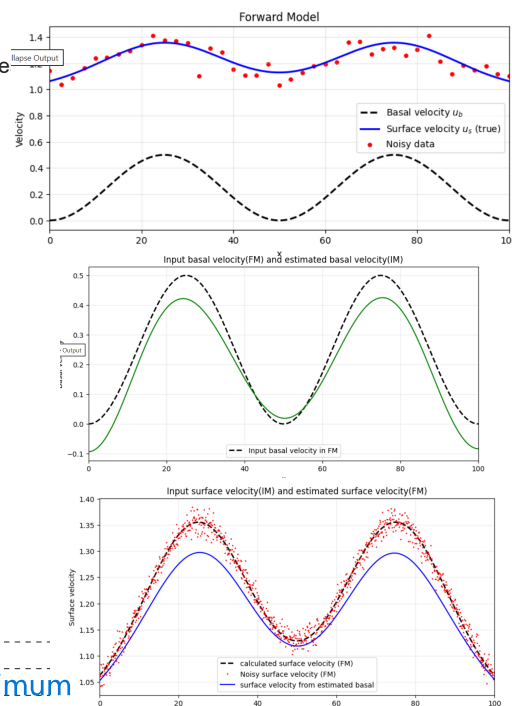
Here we compare the input/true basal velocity and the one estimated from the inverse model using first-order Tikonov regularization. Notice the higher disparities in the maximum and minimum values.

Figure 3:

Here the noisy surface velocity which is an input to the inverse model is compared with the surface velocity calculated by using the estimated basal velocity in the forward model. A similar disparity at the maximum values is observed.

This is a test problem to see if the inverse model is valid to be used in the real glacier studies. The input basal velocity profile suggests two regions of fast sliding along the glacier.

Figure



Limitations / Future Scope:

It is an idealized 1D model, similar approach can be applied to 2D or full Stokes flow model of a glacier. The further steps as given in the reference paper is to find the model/data resolution matrices, and then test to more glaciers where basal velocity data is available.

References:

Truffer (2004)
<https://www.cambridge.org/core/journals/journal-of-glaciology/article/basal-speed-of-valley-glaciers-an-inverse-approach/F0AC6A417824E6284FE00EC89BB60023#R3>
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